

Fatema Haque

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Software Engineer | Distributed Systems | Simulation Software | Networking

PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience developing interactive educational games, performance-sensitive simulation software, and distributed systems using C++, C#, and Unity. Skilled in software architecture, networking, runtime optimization, debugging, and translating complex technical requirements into reliable user-facing software. Strong foundation in mathematics and systems design, with a proven ability to deliver scalable solutions across educational technology and marine simulation domains.

CAREER HIGHLIGHTS

- Led software development of a proprietary radar DLL for an embedded navigation simulator supporting a high-fidelity naval defence training project for an international defence customer.
- Developed real-time marine simulation systems involving large-vessel bridge simulation, embedded navigation, sensor device models, radar systems, and distributed architecture.
- Contributed to a portfolio of 30+ educational games supporting blended learning in classrooms across North America, reaching 2M+ students and educators.
- Built a custom 2D game engine using C++ and SFML.
- Achieved 2nd Place at Hacking Mount Pearl 2021 for developing "Find My Way", a mentorship app connecting students to career pathways.

PROFESSIONAL EXPERIENCE

Software Developer

Virtual Marine | Paradise, NL, Canada | Aug 2025 - Present

- Developed software using C++, C#, and Unity for marine training simulation systems, with a focus on large-vessel bridge simulation, embedded navigation simulation, sensor models and radar systems.
- Led development of a proprietary radar DLL, integrating hardware interfaces, distributed communication, radar and sensor protocol handling, TCP control connections, and UDP ASTERIX CAT-240 raw video output.
- Contributed to the mathematical and physics-based modelling of 5 marine vessels, improving hydrodynamic model accuracy by approximately 20%.
- Applied numerical methods to solve real-time physics simulation problems related to vessel behaviour and model validation.
- Implemented protocol support for marine sensor models, including NMEA 0183 messaging for AIS simulation.
- Developed automation scripts to streamline software installation and networking setup, reducing commercial delivery time by approximately 30%.
- Contributed to real-time simulation components that require reliable communication between multiple software modules, external interfaces, and hardware-integrated systems.
- Configured and analyzed data flow and communication between multiple servers using Wireshark, supporting real-time communication across distributed systems.
- Produced navigational chart data and high-resolution radar imagery for multiple global simulation environments using Global Mapper, GIMP, and ENCX SDK.
- Validated radar and sensor simulation workflows by investigating message-level communication, network traffic, configuration issues, and expected system behaviour.
- Collaborated with engineers and subject matter experts to translate vessel navigation, radar, and sensor requirements into practical software implementations.

Game Developer

Carnegie Learning | St. John's, NL, Canada | Apr 2023 - Jun 2025

- Developed 30+ gamified math learning models in Unity and C#, improving engagement and learning outcomes.
- Streamlined content pipeline from Google Sheets, reducing level load times by 20%.
- Implemented math-focused simulations to teach complex mathematical concepts through interactive, system-driven gameplay.
- Leveraged AI techniques and data-driven iteration to implement adaptive learning methods, optimizing gameplay to better support diverse learners.
- Contributed to team code reviews and promoted clean software architecture practices in Unity's ECS and DOTS development workflows.
- Mentored 2 junior developers, conducting code reviews and coaching on topics including software architecture and modular design.
- Partnered with artists, designers, and educators to ensure gameplay was engaging and aligned with instructional design guidelines.

Junior Game Developer

Carnegie Learning | St. John's, NL, Canada | May 2022 - Apr 2023

- Developed comprehensive unit tests to validate core mechanics and maintain code reliability, reducing post-launch bugs by 20%.
- Improved runtime performance by profiling rendering bottlenecks and optimizing update cycles.
- Collaborated with senior developers to prototype and refine learning mechanics using 2D/3D geometric manipulation, contributing to 6 successful feature launches.
- Developed gameplay systems involving object interactions and movement physics, contributing to realistic, simulation-style player experiences.

Game Development Intern

Carnegie Learning | St. John's, NL, Canada | Jan 2021 - Apr 2022

- Recognized by senior developers for rapid onboarding, adaptability, and collaborative spirit, praised as a quick learner and positive contributor to both team culture and product quality.
- Eliminated high-impact bugs, resulting in a 15% improvement in game stability and a smoother user experience.
- Identified and addressed accessibility gaps, delivering features that enhanced usability for diverse learners.

EDUCATION

Bachelor of Science (BS): Computer Science

Memorial University of Newfoundland | St. John's, NL | 2022

KEY SKILLS

- **Programming Languages:** C#, C++, JavaScript.
- **Frameworks/Tools:** Unity, Git, PlasticSCM, TortoiseSVN, ENCX SDK, Global Mapper, GIMP, Qt, Wireshark, VMware Workstation Pro.
- **Game Development:** AI Systems, Gameplay Systems, Performance Optimization, Animation Trees, Spine Animations, ECS, DOTS.
- **Web Development:** HTML, CSS, Node.js, MySQL, Bootstrap.
- **Software Development:** Object-Oriented Programming (OOP), Entity Component System (ECS), DOTS, Software Architecture, High-Level Architecture (HLA), Distributed Systems, Networking, Design Patterns, Debugging, Systems Development, Data Structures & Algorithms.
- **Soft Skills:** Communication, Problem Solving, Mentorship, Ownership, Autonomy.
- **Cloud:** AWS Cloud Practitioner Essentials.